

smart, there exist communication protocols. And since you must spend quite a bit of money to get your home automated, it'll be worth spending some time learning about the popular protocols and their advantages and disadvantages.

Get the lowdown right here:

X10

X10 is a greybeard among communication protocols – in that it has been around for a seriously long time. The protocol was introduced in 1975 and has been used by home automation enthusiasts across the world since then.

Many of the technologies that came later would adopt the method in which X10 functioned- by utilizing the existing powerline in the home for communication. Using the protocol, one can easily control utilities such as lamps and household appliances. Not just that, you can even monitor doors and windows and detect motion, not to mention the ability to replace wall switches.



Old hand, but still has good things for you

To get geekier about how X10 works, it sends bursts of 120 kHz(RF) signals. These bursts represent digital information. The signal is sent to the powerline at the zero crossing of the AC sine wave. This is so that the

interference can be minimized. You can assign every X10 product with a house code as well as a unit code – that's up to 256 unique addresses.

UPB

UPB is also a protocol that relies on wired technology. Expanded as Universal Powerline Bus, it is similar in many ways to X10. In fact, UPB was introduced with the hope that it would replace X10. The probable reason? Because it provided a higher reliability and range. The higher reliability comes from the fact that it's less susceptible to powerline noise. The transmission range is more than 1.6 kilometers.

Since the technology makes use of the existing power lines in the home, it cuts down on the cost of sending the signals which control the devices, regardless of whether they are inside or outside the house.